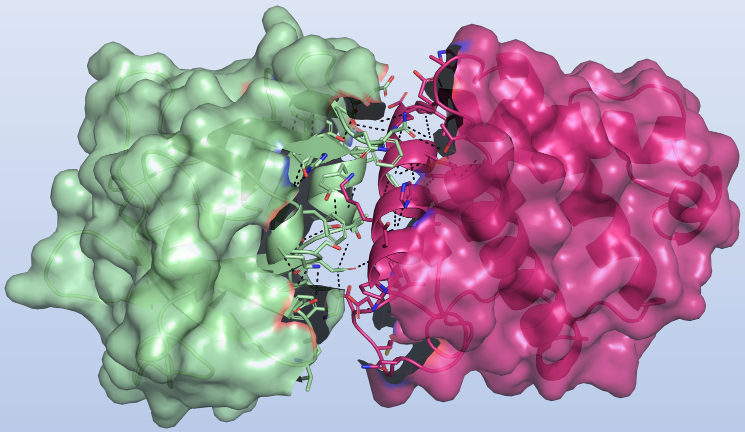
Day4: Protein Structures
Visualization of Protein Interactions

Åbo Akademi University



Prof Tiina A. Salminen

Structural Bioinformatics Laboratory

InFLAMES Research Flagship Center

Biochemistry, Faculty of Science and Engineering

Pre-assignments:

1. Install Pymol
2. Protein interactions
3. Biomolecular structures
4. AlphaFold for predicting 3D structures for proteins

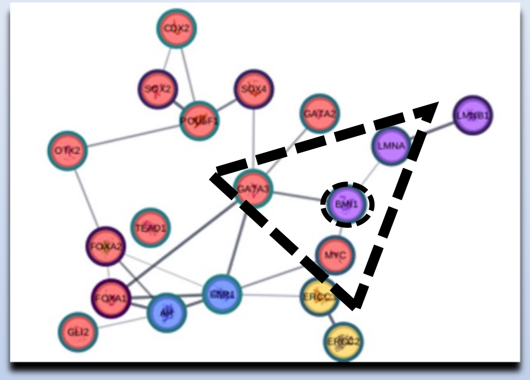
On-site:

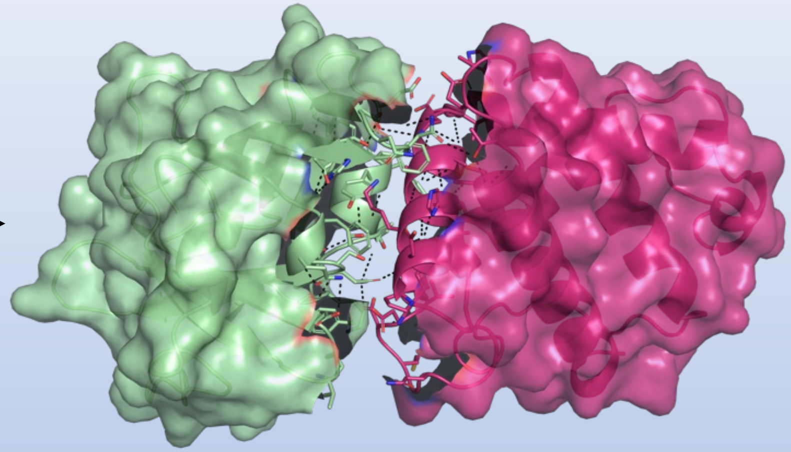
Lecture: Introduction to visualization

Exercises:

1. Individual exercise -> Learn how to use Pymol
2. Group exercise -> Presentation and project report

Day4: Protein Structures





From molecular to atomic level

<u>Outline & breaks:</u>

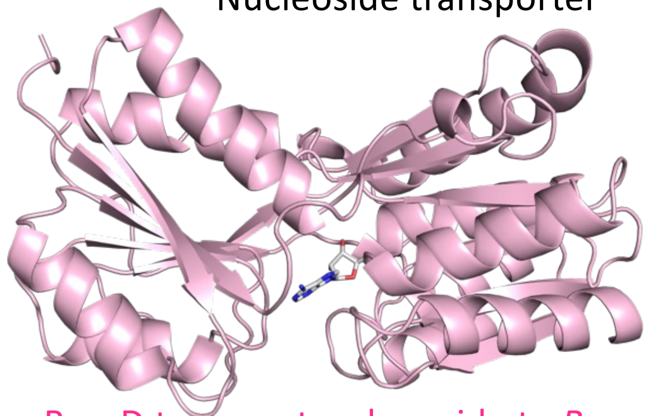
1. Introduction
   - 5 min break
2. Pymol Exercise - everyone
3. Group meeting & work
   - ”group” breaks

Why do we want to study protein structures?

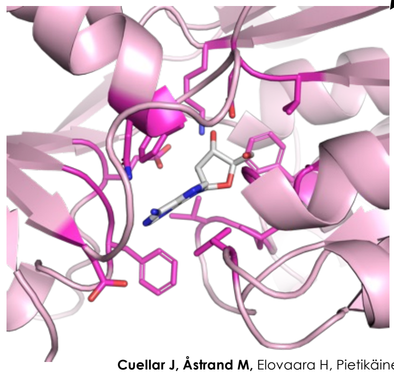
- **The function of a protein is determined by its structure!**
- If we know the structure, we can understand protein function
- Examples:
  - Drug design → therapeutic agents can be designed
  - Chemical industry → enzymes can be made more efficent
  - Agriculture → modified proteins may increase tolerance and yields
  - In your project → analyze interactions and understand function of proteins

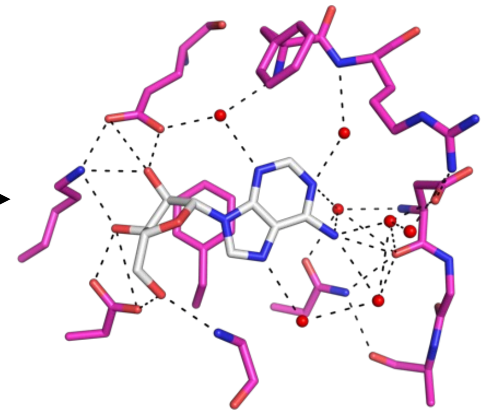
Borrelia BmpD

Nucleoside transporter



BmpD transports adenoside to *Borrelia*





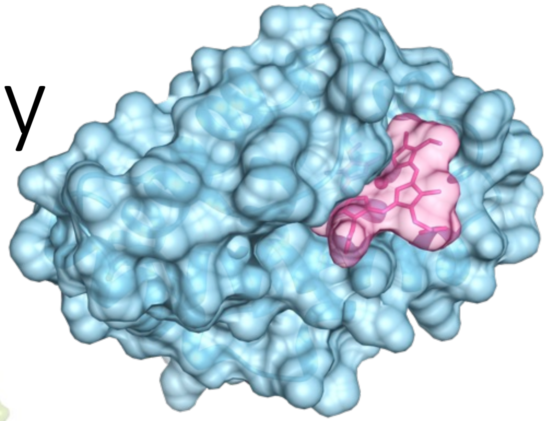
Typically, ~10% of the residues are functionally important while 90% create the 3D structure essential for function (structurally important)

Instagram #åaforskaren Mia Åstrand

Åstrand M, Cuellar J, Hytönen J, Salminen TA (2019) Predicting the ligand-binding properties of Borrelia burgdorferi s.s. Bmp proteins in light of the conserved features of related Borrelia proteins. *Journal of Theoretical Biology* 462:97-108.

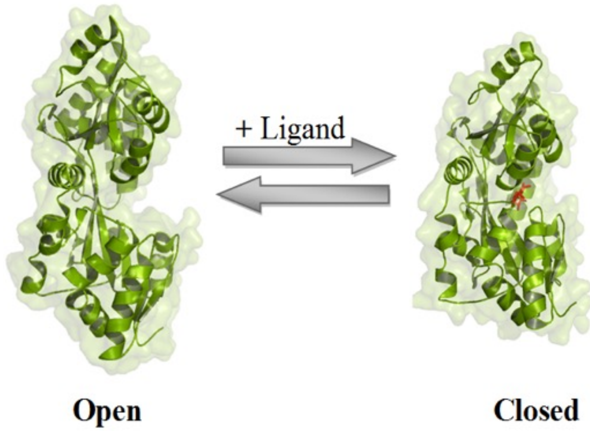
Cuellar J, Åstrand M, Elovaara H, Pietikäinen A, Sirén S, Liljeblad A, Guédez G*, Salminen TA*, Hytönen J.* (2020) Structural and biomolecular analyses of *Borrelia burgdorferi* BmpD reveal a substrate-binding protein of an ABC-type nucleoside transporter family. *Infection and Immunity* 88(4):e00962-19.

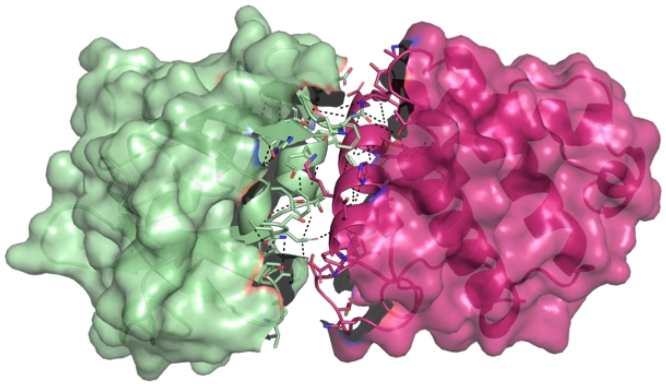
Structure is a tool to understand biology

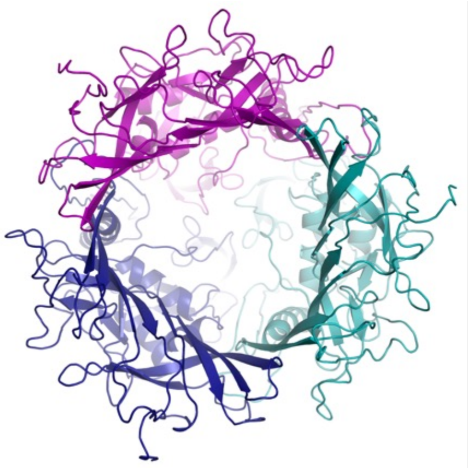
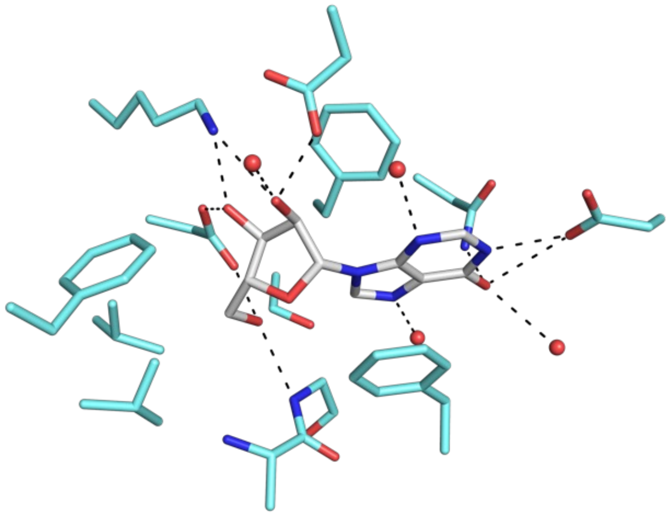
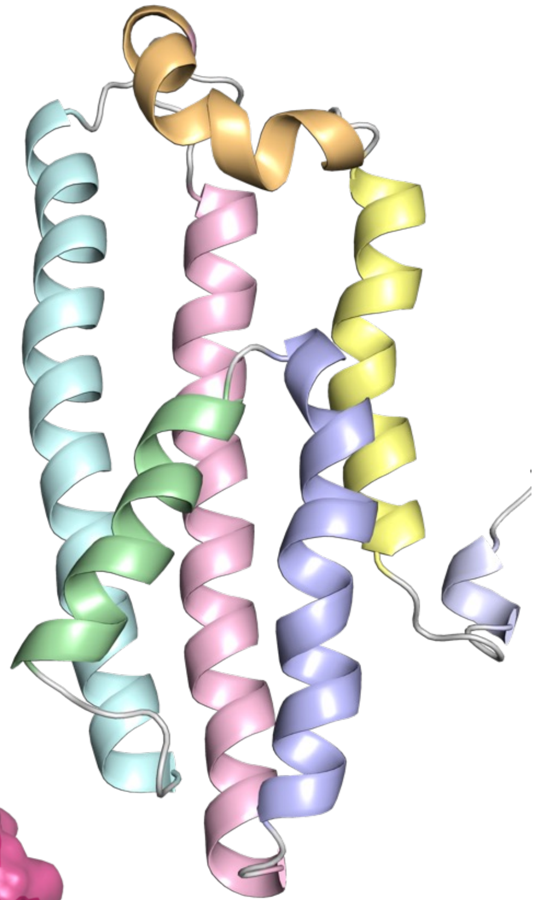


In addition to the protein, we would like to see:

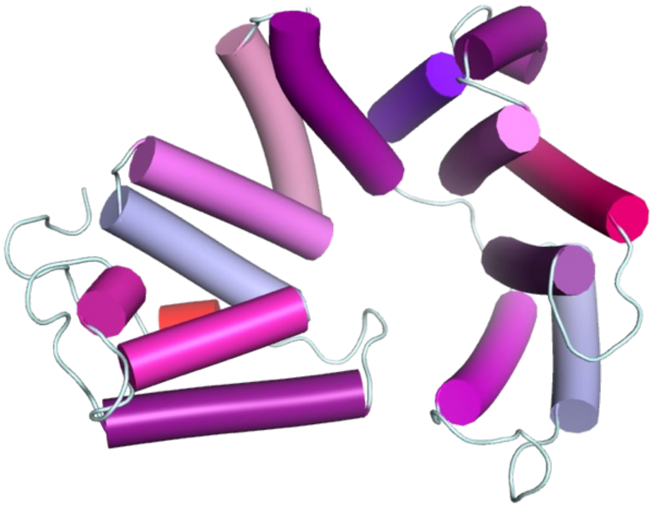
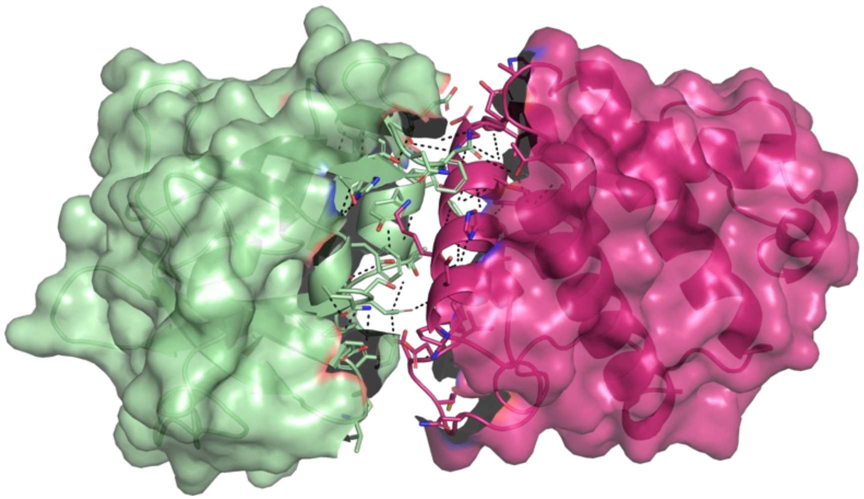
- Ligands, cofactors
- Functional conformations, protein dynamics
- Molecular interactions
    - Protein-Small molecule
    - Protein-Protein (PPI)
    - Protein-DNA
    - ➢ Complex structures





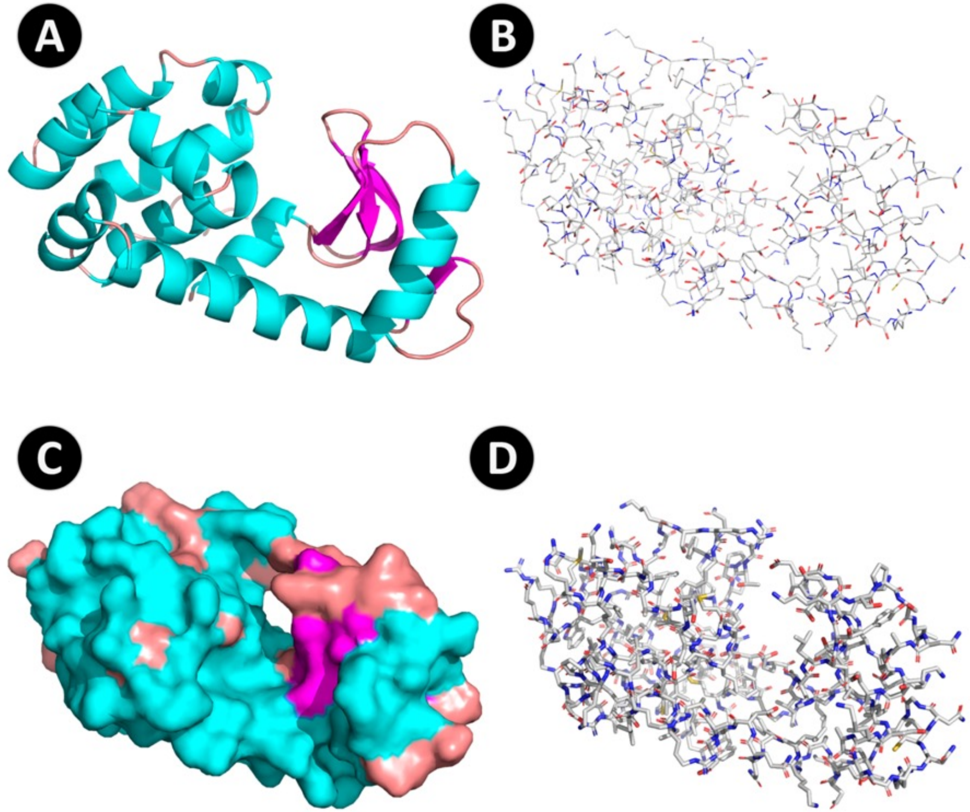




Visualization of 3D structures

Why do we need visualization?

- We depend on visual information!
- Visualization programs enable us to
    - Understand structures in 3D
    - Study protein interactions in detail
    - Understand the function of a protein
    - Prepare descriptive figures



3D structures from PDB

RCSB PDB
PROTEIN DATA BANK

- Includes 3D structures from macromolecules
  - Proteins, DNA, RNA
- Established in 1971 at Brookhaven National Laboratory
  - Contained 7 structures!
  - Helped to store and distribute structural data
- Structures have been determined using
  - X-ray crystallography
  - Nuclear Magnetic Resonance
  - Electron microscopy (cryo-EM)

CRYSTALLOGRAPHY

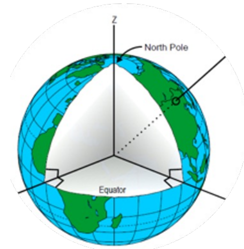
Protein Data Bank

A repository system for protein crystallographic data will be operated jointly by the Crystallographic Data Centre, Cambridge, and the Brookhaven National Laboratory. The system will be responsible for storing atomic coordinates, structure factors and electron density maps and will make these data available on request. Distribution will be on magnetic tape in machine-readable form whenever possible. There

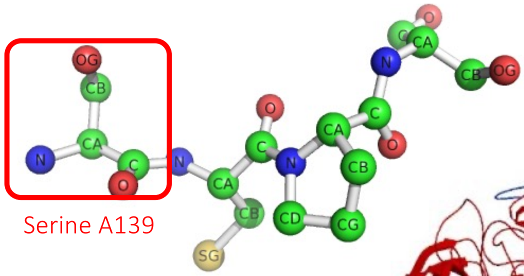
Crystallography: Protein Data Bank. *Nature New Biology* **233**, 223 (1971). https://doi.org/10.1038/newbio233223b0

The PDB File

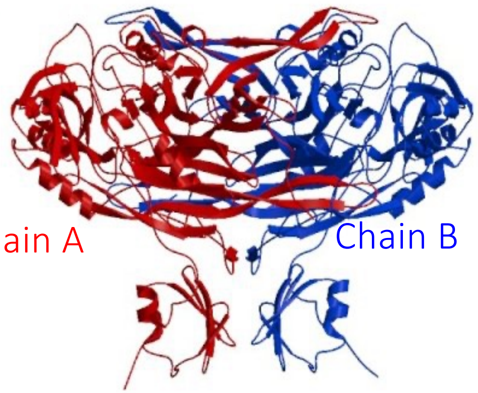
Atomic coordinates describe the location in space for each atom in the protein



		Atom	Amino acid	Chain identifier	Residue number	X	Y	Z	Occupancy	B-factor	
ATOM	1	N	SER	A	139	22.479	-4.111	18.945	1.00	34.34	N
ATOM	2	CA	SER	A	139	22.123	-2.660	18.772	1.00	36.15	C
ATOM	3	C	SER	A	139	21.917	-2.021	20.148	1.00	35.23	C
ATOM	4	O	SER	A	139	21.829	-2.724	21.148	1.00	37.66	O
ATOM	5	CB	SER	A	139	20.891	-2.485	17.863	1.00	36.40	C
ATOM	6	OG	SER	A	139	19.742	-3.154	18.362	1.00	43.71	O
ATOM	7	N	CYS	A	140	21.832	-0.699	20.188	1.00	34.49	N
ATOM	8	CA	CYS	A	140	21.716	0.023	21.450	1.00	36.17	C
ATOM	9	C	CYS	A	140	20.371	0.642	21.738	1.00	37.47	C
ATOM	10	O	CYS	A	140	19.731	1.189	20.844	1.00	36.62	O
ATOM	11	CB	CYS	A	140	22.749	1.139	21.490	1.00	35.87	C
ATOM	12	SG	CYS	A	140	24.467	0.525	21.122	1.00	36.23	S
ATOM	13	N	PRO	A	141	20.006	0.713	23.028	1.00	38.50	N
ATOM	14	CA	PRO	A	141	18.735	1.279	23.481	1.00	38.39	C
ATOM	15	C	PRO	A	141	18.517	2.743	23.086	1.00	39.20	C
ATOM	16	O	PRO	A	141	19.407	3.590	23.232	1.00	38.41	O
ATOM	17	CB	PRO	A	141	18.809	1.093	25.001	1.00	39.58	C
ATOM	18	CG	PRO	A	141	20.284	1.177	25.286	1.00	38.94	C
ATOM	19	CD	PRO	A	141	20.849	0.333	24.181	1.00	39.16	C
ATOM	20	N	SER	A	142	17.342	3.006	22.515	1.00	37.85	N
ATOM	21	CA	SER	A	142	16.939	4.352	22.097	1.00	39.22	C
ATOM	22	C	SER	A	142	17.444	4.856	20.744	1.00	34.22	C
ATOM	23	O	SER	A	142	17.045	5.945	20.316	1.00	31.72	O
ATOM	24	CB	SER	A	142	17.288	5.385	23.185	1.00	40.67	C
ATOM	25	OG	SER	A	142	16.498	5.192	24.346	1.00	48.68	O



Serine A139



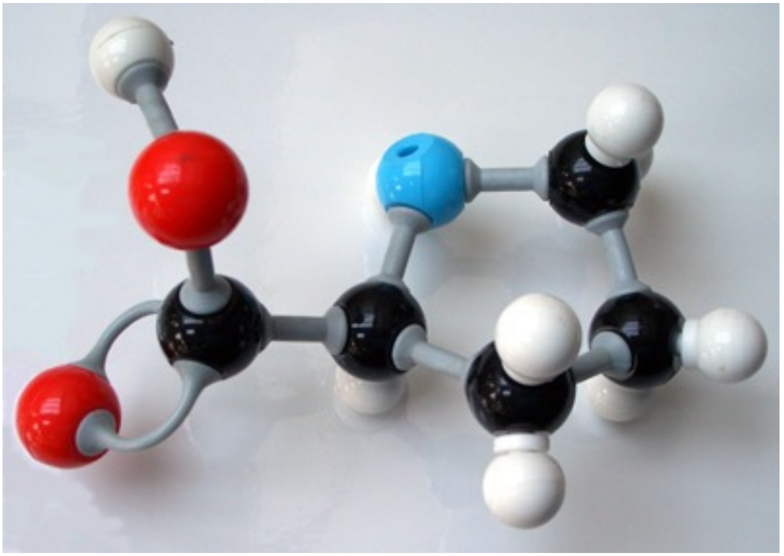
Homodimer

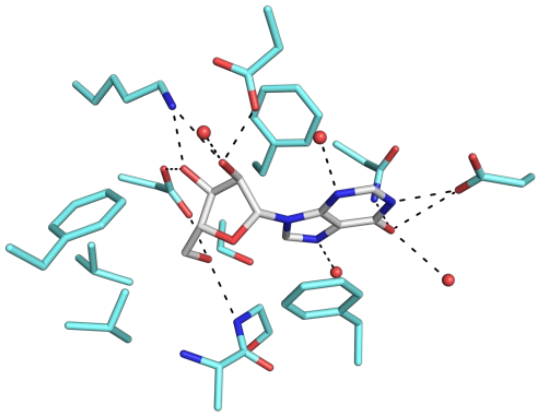
PDB Identification Code (PDB ID), unique for each structure, is a **4-character alphanumeric identifier** e.g. 1AOX

Visualization styles: Colouring

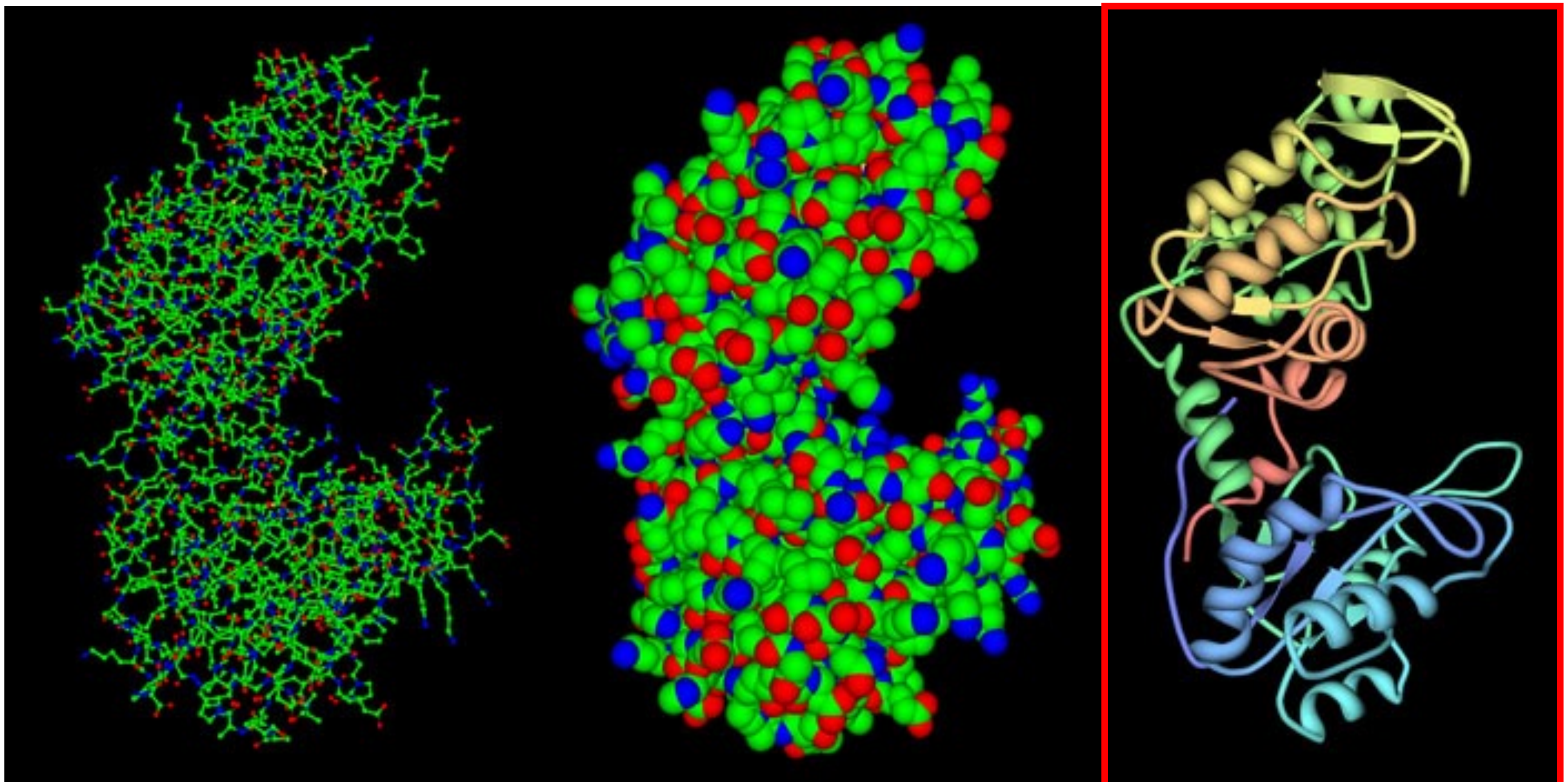
- Typical colour convention for most common atoms
  - **Black** for carbon
  - Blue for nitrogen
  - Red for oxygen
  - White for hydrogen
  - Deep yellow for sulphur
  - Purple for phosphorus
  - Light, medium, medium dark, and dark green for the halogens (F, Cl, Br, I)
  - Silver for metals (Co, Fe, Ni, Cu)

A plastic ball-and-stick model of proline





Visualization styles



Pdb-101

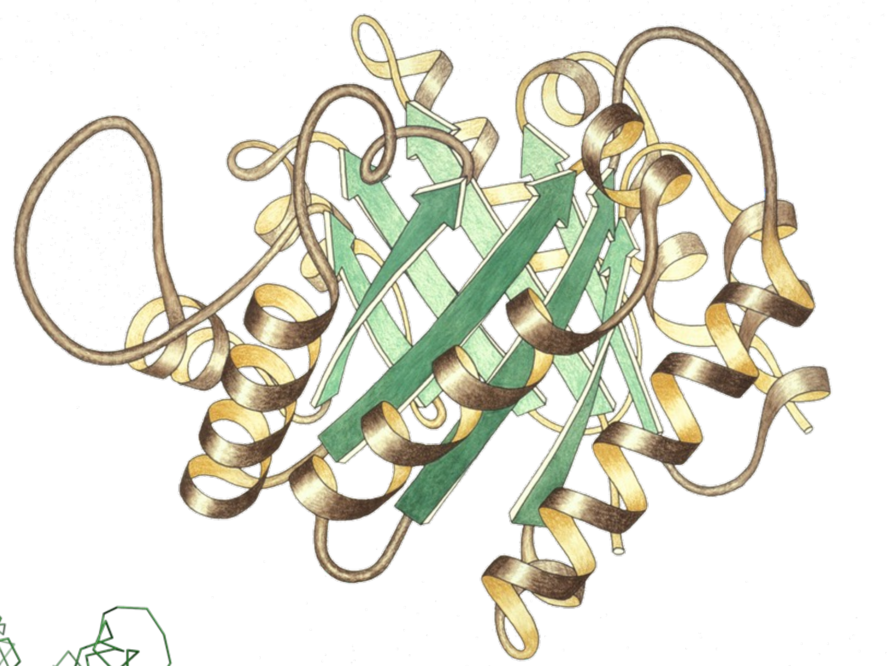
Wireframe
Ball and Stick

Space filling
or
Surface

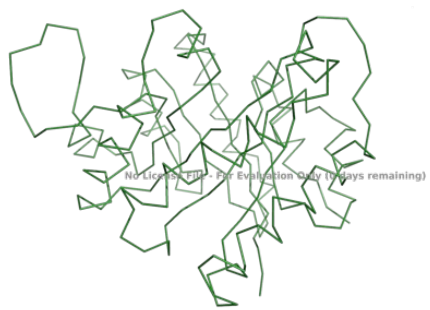
Cartoon
Ribbon diagram

Visualization styles in Pymol: Cartoon or ribbon

- Highlights secondary structures and folds
- Cartoon draws secondary structures as arrows/helices
- Ribbon is a CA trace, which draws a line from the Cα atom on each amino acid to the next → traces the fold



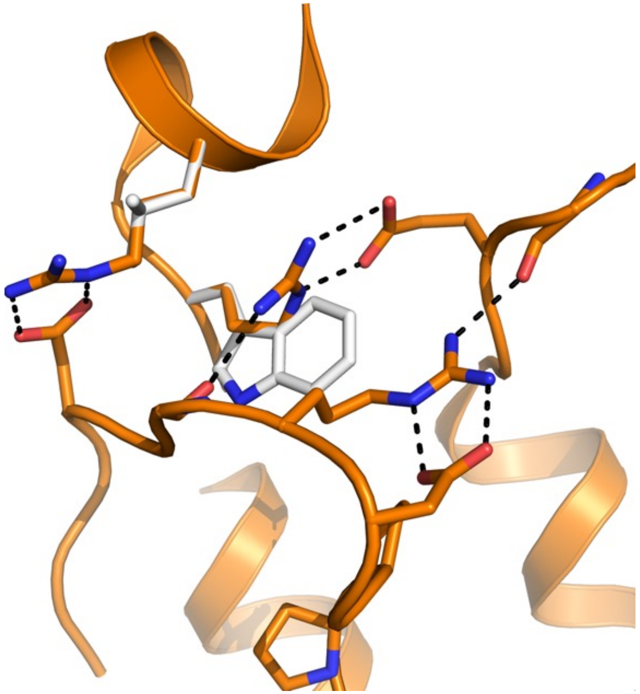
TIM-barrel
Jane Richardson
Wikimedia Commons



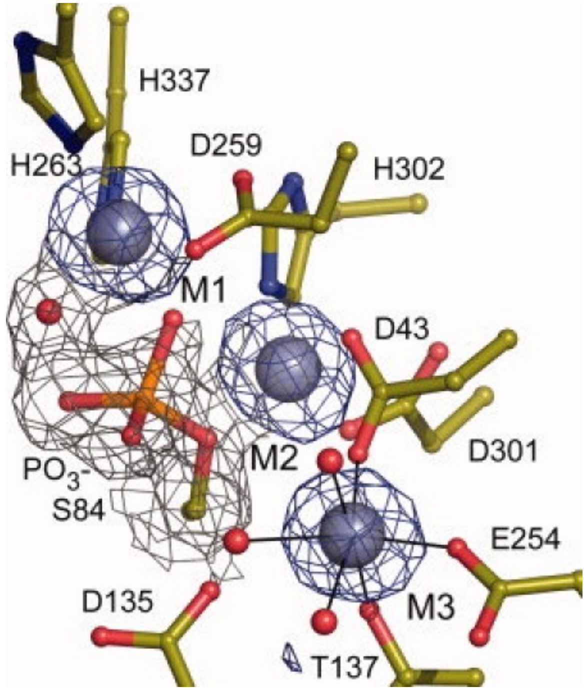
PDB ID 1TIM in ribbon

Visualization styles for atoms

- Lines, sticks and ball-and-sticks for displaying details:
    - Amino acids
    - Bonds (interactions)
    - All atoms shown in the same size!
- Waters and other atoms as spheres
    - Relative to their atom size
- Waters are often shown as non-bonded spheres (nb_spheres)

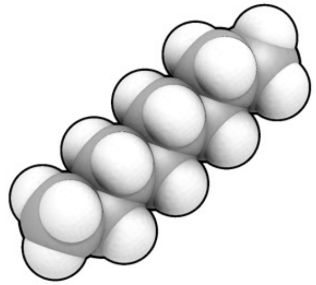


Inherited α-mannosidase mutations. Change of amino acid from R to W destroys interactions in human lysosomal α-mannosidase. Heikinheimo, unpublished

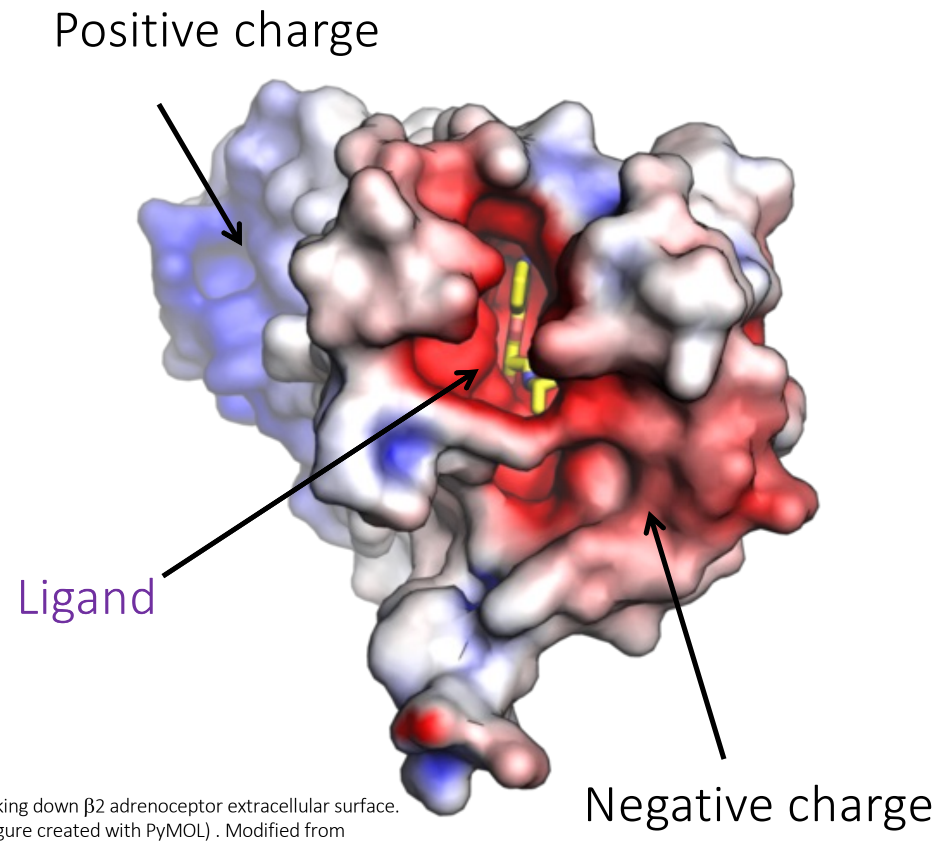


Zn^{2+} coordination and electron density map at alkaline phosphatase active site Koutsiolis 2010, doi: 10.1002/pro.284.

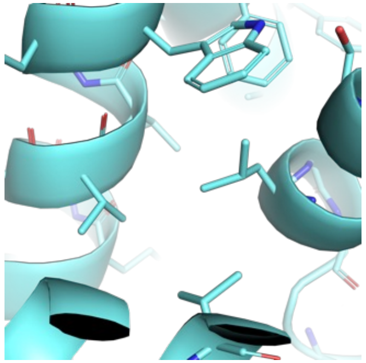
Visualization styles: Space filling and surf



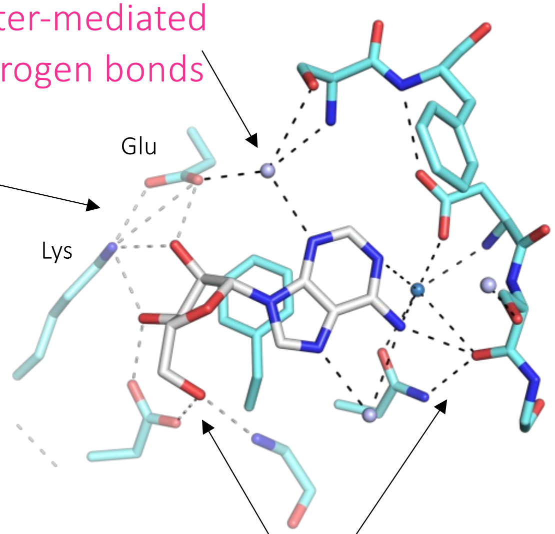
- In space-filling representation, atoms are shown as spheres
  - Radius is proportional to atom radius
- Surfaces
  - Surface charges
  - Good for displaying volume, cavities, shape etc.



Charge surface looking down β2 adrenoceptor extracellular surface. (PDB code 2RH1, figure created with PyMOL) . Modified from Opabinia regalis, Wikimedia commons, cc BY-SA 3.0



Hydrophobic interactions



Molecular interactions

1. Hydrophobic interactions

- Non-polar, hydrophobic residues pack together because they cannot participate in hydrogen bonding with water
  - Inside protein
  - Protein-protein interfaces

1. Polar interactions

- Hydrogen bonds
  - Two polar groups interact: O-H:::O; N-H:::O; N-H:::N; O-H:::N
- Ionic interactions (Salt bridges)
  - Positively charged Arg or Lys (or His) with negatively charged Glu or Asp

1. Covalent interactions

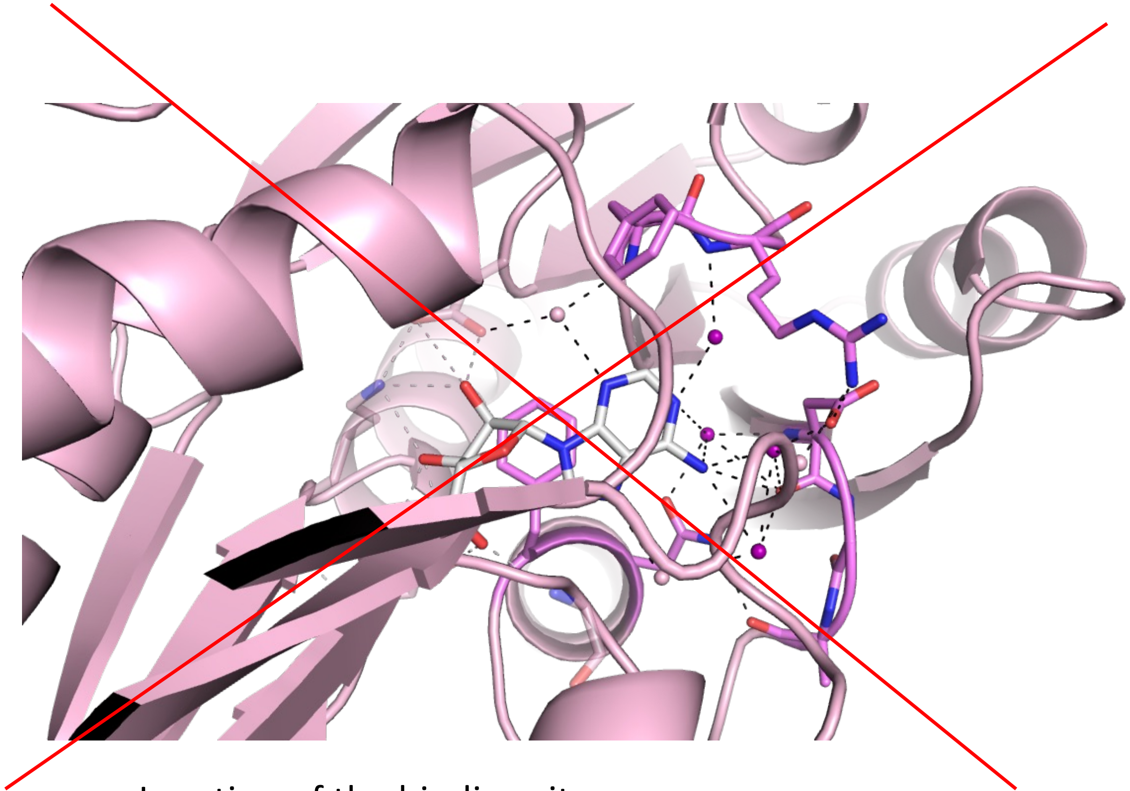
- Disulphide bonds (formed between two cysteine residues)
- Metal ions e.g. Mg^{2+}, Mn^{2+}, Cu^{2+}, Zn^{2+}

How to make a good protein figure:

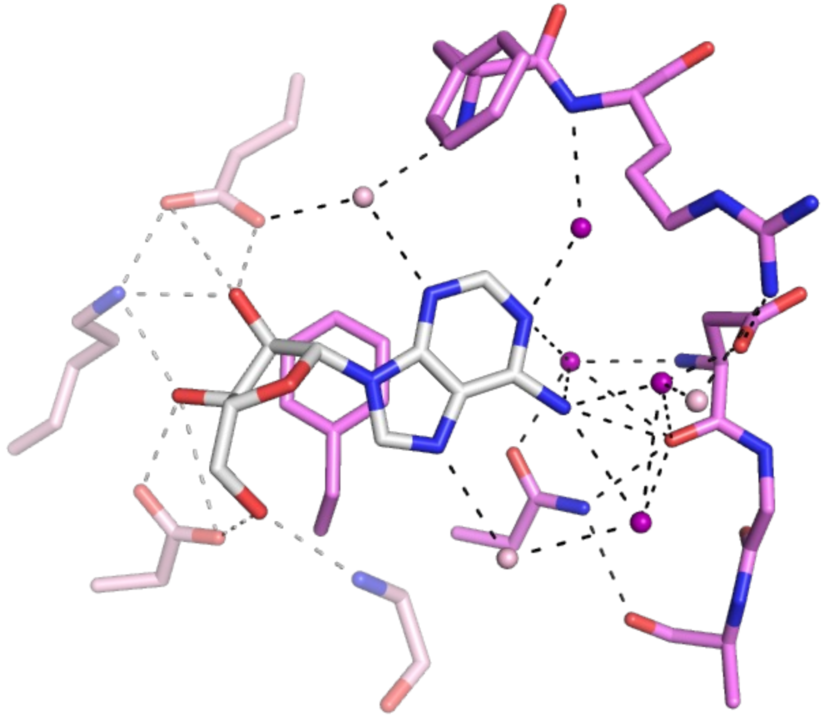
1. Make it clear and simple!
    - Highlight the most important things and remove distractions
2. Use colors, labels, arrows etc. wisely
    - Select contrasting colors to highlight differences
    - Label the most important things (e.g. catalytic residues, secondary structures involved in interactions)
3. Choose a format that is suitable for what you want to show
    - E.g. cartoon, sticks and surface representations can be used to highlight differents aspects of a protein
4. Write a clear and descriptive figure legend!
    - All colors, signs etc. should be clearly explained
    - PDB structures used should be referenced

Example: Showing ligand-bindning interactions

Focus on your message!



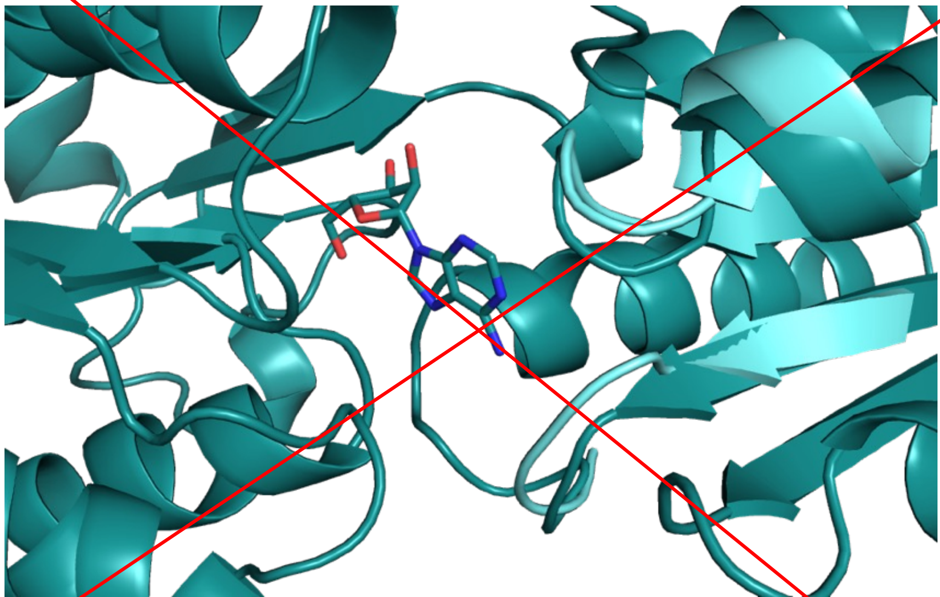
Location of the binding site



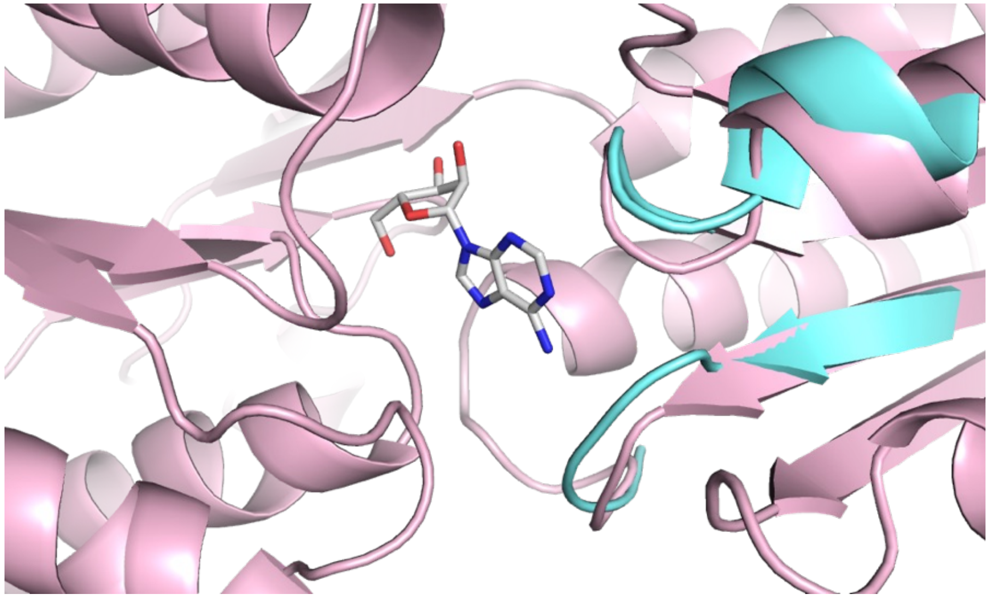
Interactions between ligand (white) and protein (pink)

Example: Coloring two superimposed structures

Colors do matter!



The two colors are too similar to show differences without and with ligand



The structure without ligand (cyan) and with ligand (pink) -> The differences are clearly visible!

Example: Figure and legend

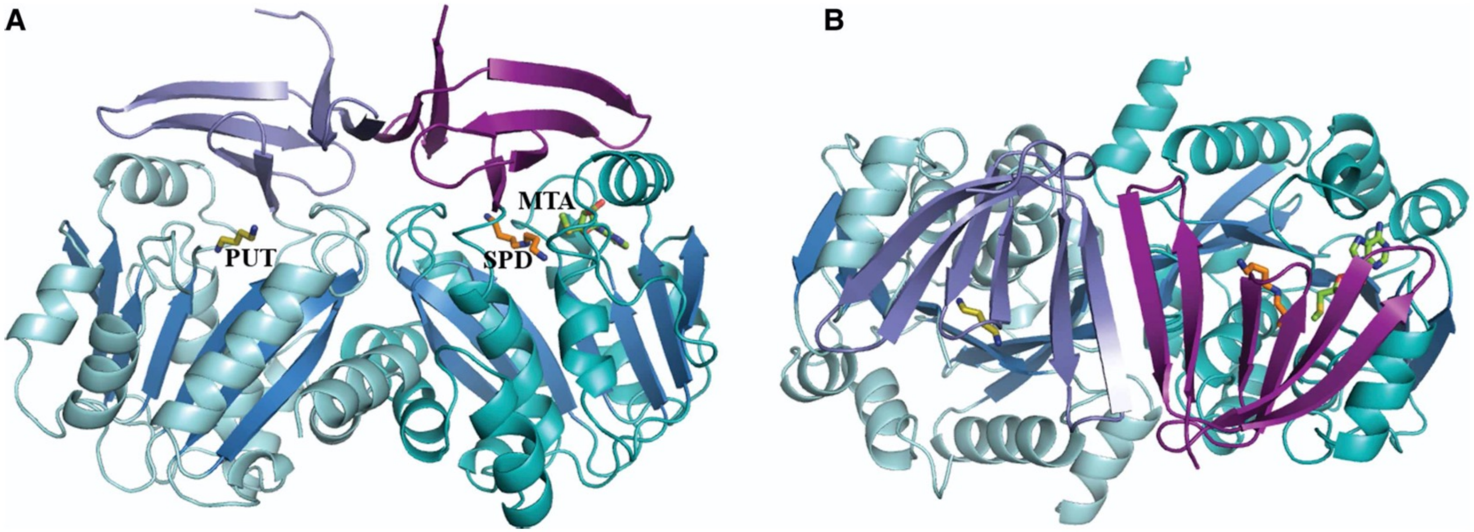
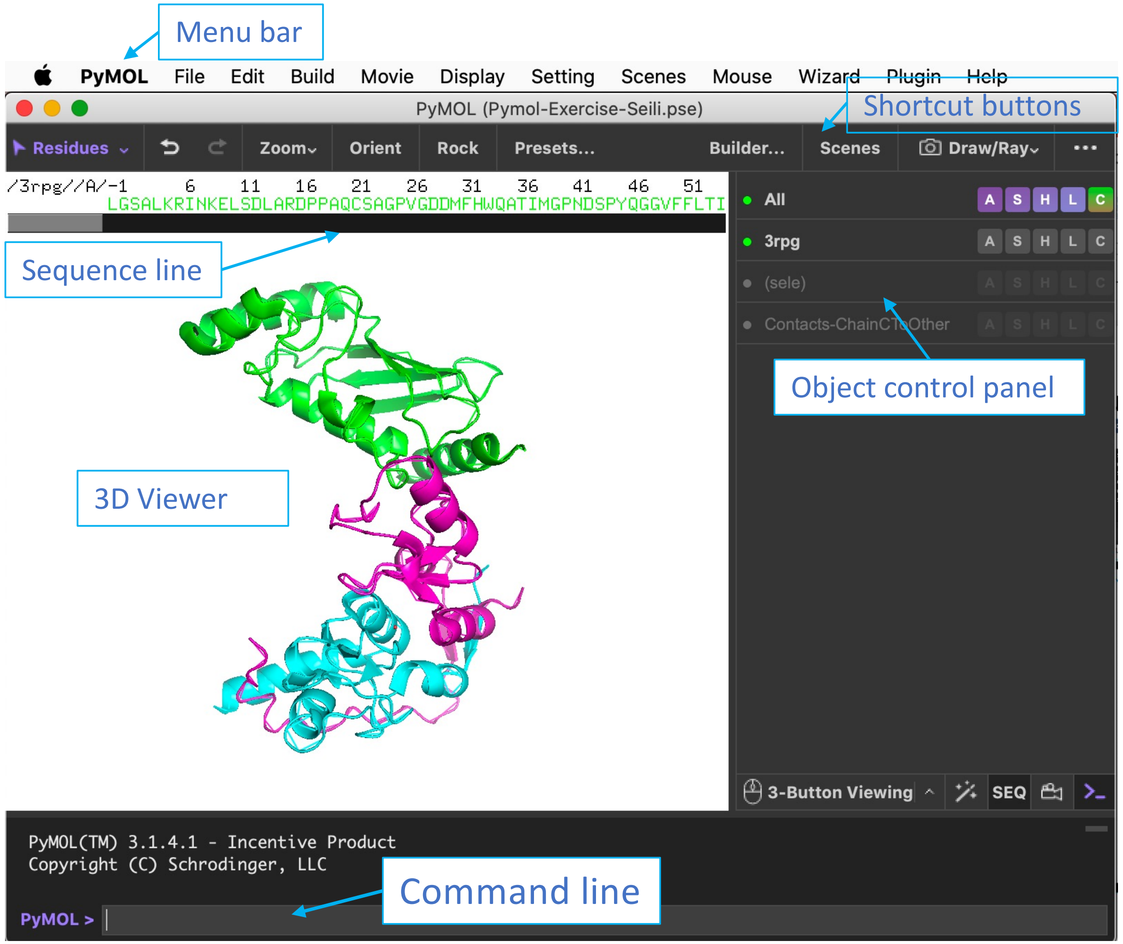


Figure 2. Structure of *Sy*SPDS with bound substrate and product.
Side (**A**) and top (**B**) view of the SPDS dimer with the N-terminal β-stranded domain of each monomer shown in light and dark purple, respectively and the C-terminal Rossmann fold-like domain with the β-stranded core (blue) shown in cyan and turquoise, respectively. Putrescine (PUT, olive), spermidine (SPD, orange) and 5′-methylthioadenosine (MTA, green) are shown in sticks.

PyMOL - The program we will use!

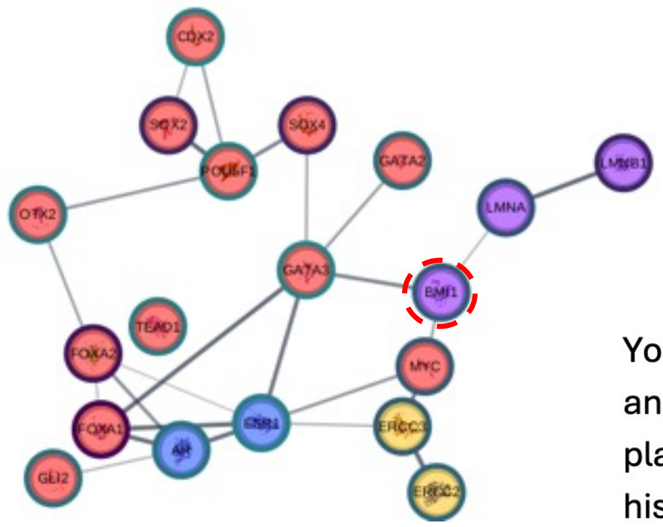
- Created by Warren DeLano
  - First release in 2000
  - Python script, OpenGL, open source
  - Since 2010 supported and distributed by Schrödinger, Inc.



- **Possibility to save sessions – Do it!**
- High quality rendering, movies etc.
- Automatic retrieval of molecules from PDB
- Possibility to use scripts to do complicated tasks
- Allow user to fully control and adjust the items displayed

https://en.wikipedia.org/wiki/PyMOL

Exercise – To get familiar with Pymol

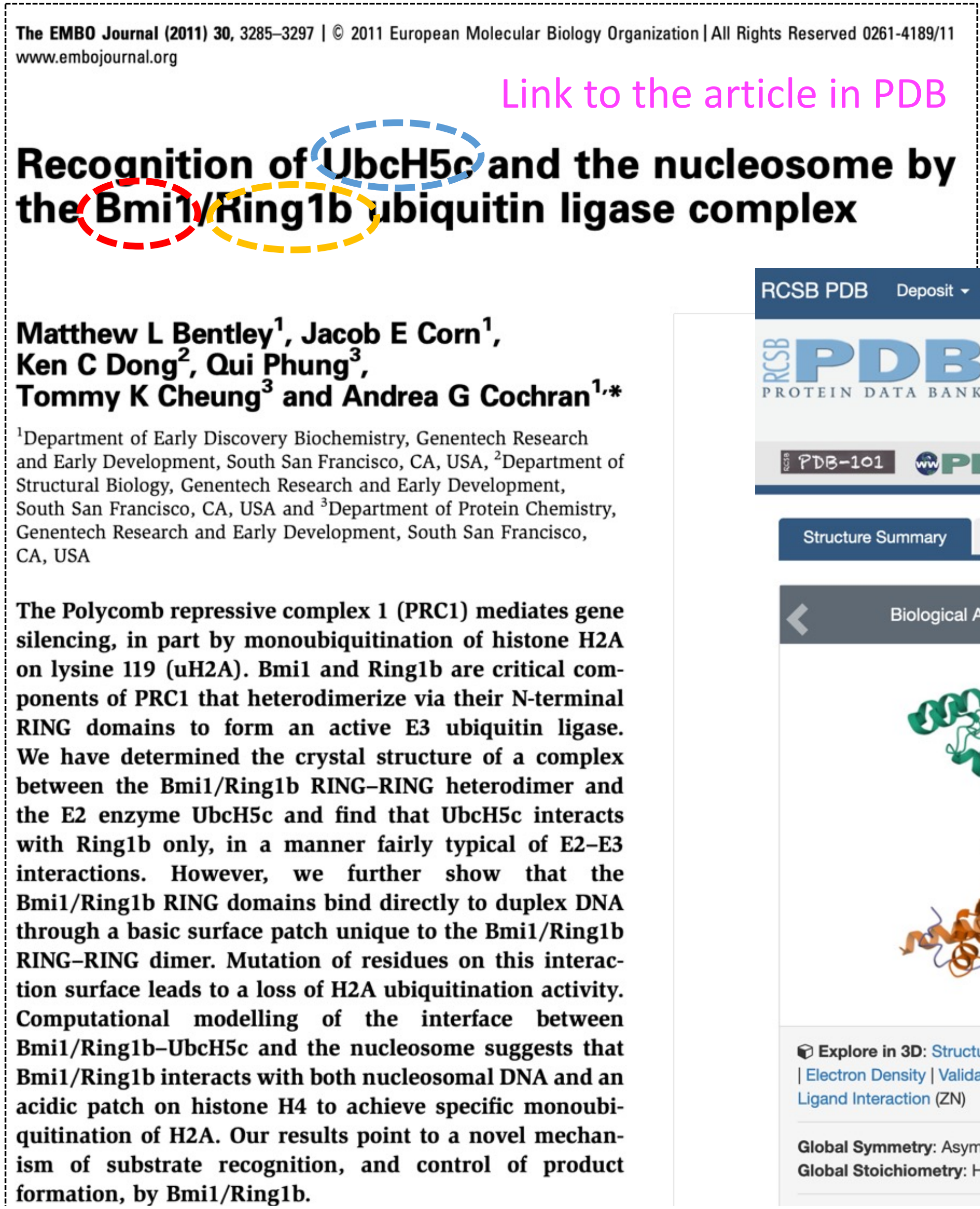


PyMOL exercise

Bmi1/Ring1b-UbcH5c complex structure

You will look at the 3D structures of three interacting proteins including RING2 (chain C) and BMI1 (chain B), both being part of human polycomb repressive complex 1-like, which plays a role in histone and gene regulation as describe by Wang et al. (2004) in “Role of histone H2A ubiquitination in polycomb silencing”. These two proteins interact as a heterodimer with the E2 enzyme UbcH5c (chain A) and the complex structure has been crystallized by Bentley et al. (2011) “Recognition of UbcH5c and the nucleosome by the BMI1/RING1b ubiquitin ligase complex” (PDB ID 3RPG). Now you will analyze how these proteins interact with each other.

PDB Search for the structure



The EMBO Journal (2011) 30, 3285–3297 | © 2011 European Molecular Biology Organization | All Rights Reserved 0261-4189/11
www.embojournal.org

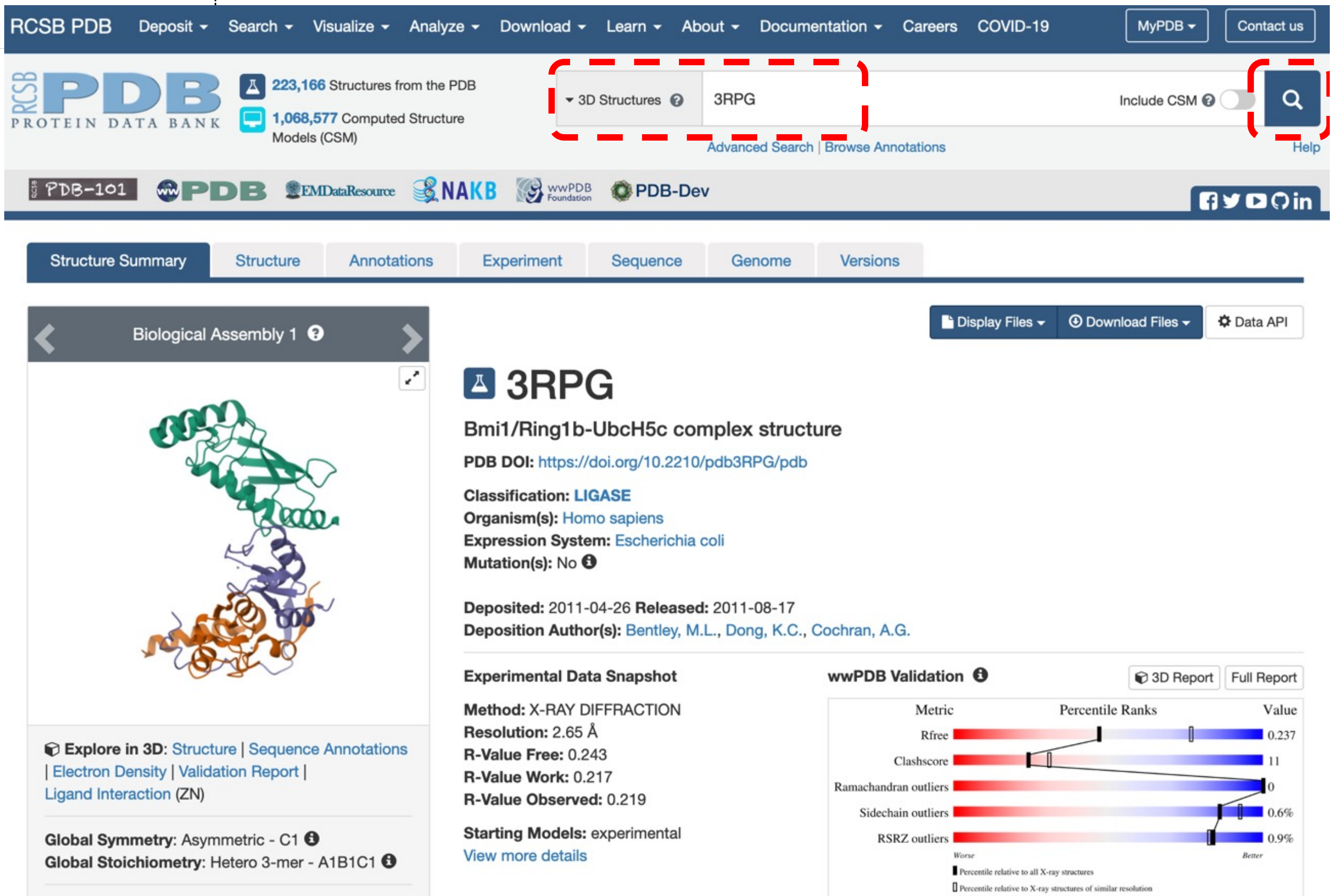
Link to the article in PDB

Recognition of UbcH5c and the nucleosome by the Bmi1/Ring1b ubiquitin ligase complex

Matthew L Bentley[1], Jacob E Corn[1], Ken C Dong[2], Qui Phung[3], Tommy K Cheung[3] and Andrea G Cochran[1,*]

[1]Department of Early Discovery Biochemistry, Genentech Research and Early Development, South San Francisco, CA, USA, [2]Department of Structural Biology, Genentech Research and Early Development, South San Francisco, CA, USA and [3]Department of Protein Chemistry, Genentech Research and Early Development, South San Francisco, CA, USA

The Polycomb repressive complex 1 (PRC1) mediates gene silencing, in part by monoubiquitination of histone H2A on lysine 119 (uH2A). Bmi1 and Ring1b are critical components of PRC1 that heterodimerize via their N-terminal RING domains to form an active E3 ubiquitin ligase. We have determined the crystal structure of a complex between the Bmi1/Ring1b RING–RING heterodimer and the E2 enzyme UbcH5c and find that UbcH5c interacts with Ring1b only, in a manner fairly typical of E2–E3 interactions. However, we further show that the Bmi1/Ring1b RING domains bind directly to duplex DNA through a basic surface patch unique to the Bmi1/Ring1b RING–RING dimer. Mutation of residues on this interaction surface leads to a loss of H2A ubiquitination activity. Computational modelling of the interface between Bmi1/Ring1b–UbcH5c and the nucleosome suggests that Bmi1/Ring1b interacts with both nucleosomal DNA and an acidic patch on histone H4 to achieve specific monoubiquitination of H2A. Our results point to a novel mechanism of substrate recognition, and control of product formation, by Bmi1/Ring1b.



PDB has information about the 3 proteins in the complex structure: chain A, B and C

Biological assembly 1 assigned by authors.

Macromolecule Content

- Total Structure Weight: 44.26 kDa
- Atom Count: 2,912
- Modelled Residue Count: 349
- Deposited Residue Count: 387
- Unique protein chains: 3

Macromolecules

Find similar proteins by: Sequence (by identity cutoff) | 3D Structure

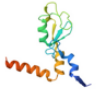
Entity ID: 1

Molecule	Chains	Sequence Length	Organism	Details	Image
Ubiquitin-conjugating enzyme E2 D3	A	149	Homo sapiens	**Mutation(s):** 0 **Gene Names:** UBE2D3, UBC5C, UBCH5C **EC:** 6.3.2.19	

UniProt & NIH Common Fund Data Resources

Find proteins for P61077 *(Homo sapiens)* Explore P61077 Go to UniProtKB: P61077

Entity ID: 2

Molecule	Chains	Sequence Length	Organism	Details	Image
Polycomb complex protein BMI-1	B	117	Homo sapiens	**Mutation(s):** 0 **Gene Names:** BMI1, PCGF4, RNF51	

UniProt & NIH Common Fund Data Resources

Find proteins for P35226 *(Homo sapiens)* Explore P35226 Go to UniProtKB: P35226

Entity ID: 3

Molecule	Chains	Sequence Length	Organism	Details	Image
E3 ubiquitin-protein ligase RING2	C	121	Homo sapiens	**Mutation(s):** 0 **Gene Names:** RNF2, BAP1, DING, HIPI3, RING1B **EC:** 6.3.2	

UniProt & NIH Common Fund Data Resources

Find proteins for Q99496 *(Homo sapiens)* Explore Q99496 Go to UniProtKB: Q99496

Literature

Download Primary Citation

Recognition of UbcH5c and the nucleosome by the Bmi1/Ring1b ubiquitin ligase complex.

Bentley, M.L., Corn, J.E., Dong, K.C., Phung, Q., Cheung, T.K., Cochran, A.G.

(2011) EMBO J **30**: 3285-3297

PubMed: 21772249 Search on PubMed Search on PubMed Central

DOI: https://doi.org/10.1038/emboj.2011.243

Primary Citation of Related Structures:

3RPG

PubMed Abstract:

The Polycomb repressive complex 1 (PRC1) mediates gene silencing, in part by monoubiquitination of histone H2A on lysine 119 (uH2A). Bmi1 and Ring1b are critical components of PRC1 that heterodimerize via their N-terminal RING domains to form an active E3 ubiquitin ligase. We have determined the crystal structure of a complex...

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Organizational Affiliation:

Department of Early Discovery Biochemistry, Genentech Research and Early Development, South San Francisco, CA, USA.

Small Molecules

Ligands 1 Unique

ID	Chains	Name / Formula / InChI Key	2D Diagram	3D Interactions
ZN Query on ZN Download Ideal Coordinates CCD File Download Instance Coordinates	D [auth B], E [auth B], F [auth C], G [auth C]	**ZINC ION** Zn PTFCDOFLOPIGGS-UHFFFAOYSA-N	Zn^{+2}	Interactions Interactions & Density

Pymol and Group exercises

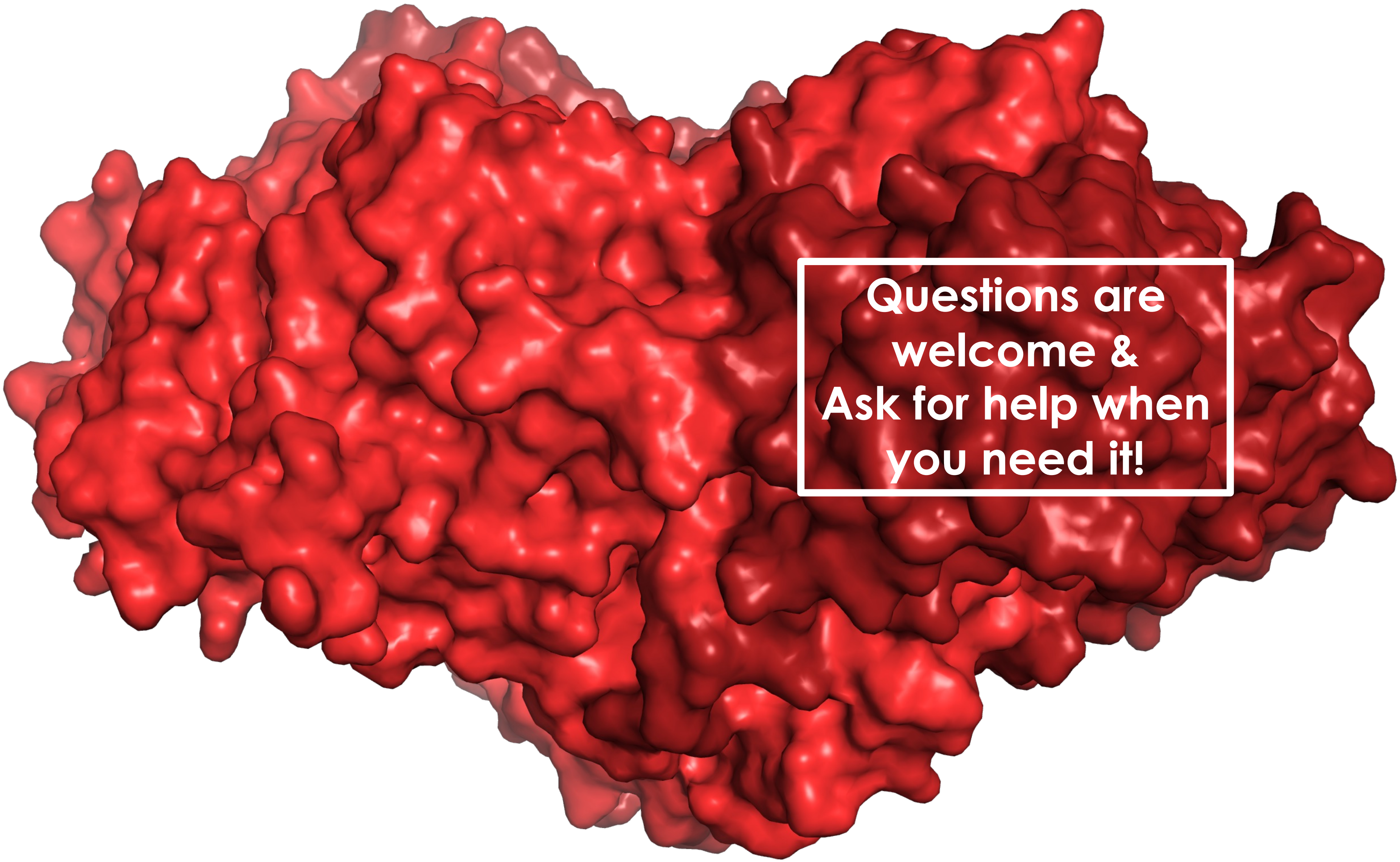
1. Individual Pymol exercise: Everyone in the group will do the exercise to learn how to use Pymol

- **Have you installed PyMol** on your computer?
- Save the license file from GitHub
- Open Pymol and read the license file in Pymol
- Follow the detailed information in the Pymol exercise instructions (GitHub)

2. Group work for the report:

In the group:

- Discuss and select one of these alternative group projects:
  1. Continue working with the same complex structure (Bentley et al. article in the GitHub)
  2. **OR** you may ask us for another protein complex to work with if you want to work e.g. DNA-binding protein
  3. OR work with the AlphaFold models you have earlier created
- Analyze the complex structure and get familiar with the proteins using the article linked in the PDB.
- Design what kind of figures and figure legends you will prepare to explain how the protein complex functions
- Submit figures and figure legends in the report


Questions are
welcome &
Ask for help when
you need it!